

Functional Requirements

Functional Requirement	Value and Unit	Type
Material To Engrave	Acrylic	N/A
Variation in Depth of Cut (Z-repeatability)	.002"	Maximum
In-Plane variation (XY-Repeatability)	.004"	Maximum
Depth of Cut	.005"	Maximum
Feed Rate	12 in/min	Maximum

- Z-repeatability
 - Due to the 30 degree angle of the router bit, the width of the cut varies with the depth of the cut. By using simple trigonometry, if the depth of cut is .005", the width of the cut should be .0027". If the depth of cut is allowed to vary to .009", the resulting width of cut is .0048", approximately .002". This amount is on the tighter end of tolerances that we can achieve with professional lathes and mills, so it is reasonable to assume that a user will not notice a cut width change of .002". However, if the variability is .004" then there is a chance that with shallower cuts the tool might not even cut the part. Therefore, a smaller desired variability should be less than the shallowest cut. A reasonable shallowest cut is .003", which is similar to what you would see on a final pass on a lathe. Therefore, a variability of .002" guarantees that a cut will actually be made.
- XY Repeatability
 - Looking at examples of things to engrave online, many of the features seem to be on the order of at least 1mm (~.04") or larger. It is reasonable to assume that a user will not notice a 10% difference in the desired feature.
- Depth of Cut
 - Consensus from online CNC communities as well as the 2.72 lecture suggest that .005" is a reasonable maximum depth of cut for engraving. http://www.2linc.com/engraving_assistance_feeds_speeds.htm
- Feed Rate
 - The recommended chip tooth load is between .003"-.006". Given that we have a maximum spindle speed of 2000RPM, we can calculate that: $FR = fpt * N * n$ where FR is the feed rate, fpt is the feed per tooth (chip load), N is the spindle speed (RPM) and n is the number of teeth on the tool bit.
 - The width of the table bed is approximately 12". With a feed rate of 12 in/min, it would take a full minute to cross the width of the table while cutting.
 - As a user, this seems a little slower than I would prefer. However, many cuts and projects will not require the span of the entire table and will in fact be perhaps simple text over a few inches. In this case, it is reasonable to assume that engraving some text or a simple shape will only take a few minutes, which is on par with that of a laser cutter.
- Discussion of XY and Z Repeatability Comparison
 - Assuming this router is meant for hobby applications, such as making signs, coasters, decorations, etc, the xy axis seems to be the most sensitive direction. It will be very obvious to the user if their desired circle comes out to be an oval or their text all melds together. For hobby applications, the z-axis is only relevant if the user is attempting to have a 3D effect. For a simple sign, the z-axis is a non-sensitive direction. However, because the engraver tool is triangular, the width of the cut varies with the depth of cut, It can be very noticeable to the user if half of the cut is done at a certain depth and the other half is done at another. Or for example, random points in the contour may seem to be 'dotted' based on depth variation. However, a quick check to determine the variation in cutting width shows that there is actually a only .002" width change with a depth change of .004". It is reasonable to assume that a user may not notice a .002" width difference. The repeatability requirements for the xy axis and z axis are coincidentally the same, but due to the previously described assumptions they are mutually exclusive.

Cutting Loads

Based on the recommended chip load, depth of cut and specific cutting energy of acrylic, the cutting forces can be estimated using the following equation:

$$F_c = u_s A$$

Where A is the cross sectional area of the chip being cut

$$A = t \cos(\phi) f$$

Where t is the depth of cut, ϕ is the cutting angle of the tool and f is the tool feed. Using the following we can find the feed

$$f = \frac{fr}{N}$$

In total:

$$F_c = u_s t \cos(\phi) \frac{fr}{N} = \frac{\left(0.3 \frac{J}{mm^3}\right) \left(\frac{1 mm^3}{0.00006 in^3}\right) (.005 in) \cos(15) \left(12 \frac{in}{min}\right) \left(\frac{1 in}{.0254 m}\right)}{\left(2000 \frac{rev}{min}\right)} = 9.64 N \sim 10 N$$

However, as this is a first order simplified approximation, it is reasonable to assume only one significant digit and round up to 10 N.

For simplicity, we can assume the cutting force is equal to the thrust force in determining structural stiffness/deflections.

Structural Required Minimum Stiffness

Axis	Maximum Deflection at tool tip	Required Stiffness at tool tip
X	0.0028in (.07mm)	1.4 E5 N/m, 140 N/mm
Y	0.0028in (.07mm)	1.4 E5 N/m, 140N/mm
Z	.002 in (.05 mm)	1.9E5 N/m, 190 N/mm

This table depicts the minimum stiffness that each axis must have when the deflection and forces are acting at the interface between the tool and the part.

Assuming that the xy plane repeatability is the vector sum of the x and y axis errors, the error apportionment between these two axes can be found using the RSS of the x and y axis errors and setting it equal to the desired repeatability of .004” from the functional requirements, yielding a maximum deflection at the tool tip of .0028 in in these axes.

The maximum deflection in the z axis directly comes from the functional requirements.

Engraver Stiffness Experiments

For a 95% confidence interval for a measurement with less than 30 samples, the following equation can be used:

$$\mu_{true} = \mu_{mean} \pm t_{.025,v} \frac{\sigma}{\sqrt{n}}$$

Where μ_{mean} is the mean of the samples taken, $t_{.025,v}$ is the scaling factor to increase the precision uncertainty, σ is the standard deviation and n is the number of samples taken. For 3 data samples, $t_{.025,v} = 3.182$.

The values represented in following experiments used this formula to compute the mean and uncertainty.

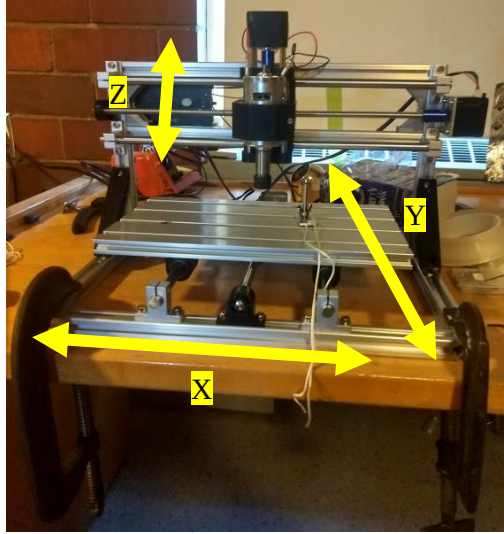


Fig 1. Experimental setup. The router is clamped to a desk

Engraver Stiffness Experiments Summary Table

Stiffness Location	Stiffness Value
Gantry Through Ground	$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} inf \\ inf \\ inf \end{bmatrix} \frac{N}{mm}$
Y-Platform	$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} 1000 \\ 300 \\ 500 \end{bmatrix} \pm \begin{bmatrix} 270 \\ 16 \\ 36 \end{bmatrix} \frac{N}{mm}$
8020 Gantry Cantilever	$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} 400 \\ 290 \\ inf \end{bmatrix} \pm \begin{bmatrix} 74 \\ 21 \\ 0 \end{bmatrix} \frac{N}{mm}$
Full Gantry	$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} 90 \\ 128 \\ 18 \end{bmatrix} \pm \begin{bmatrix} 60 \\ 8 \\ 3 \end{bmatrix} \frac{N}{mm}$
Z-Axis carriage + Bracket Stiffness	$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} 88 \\ 416 \\ 67 \end{bmatrix} \pm \begin{bmatrix} 6 \\ 10 \\ 3 \end{bmatrix} \frac{N}{mm}$
Spindle Bracket	$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} 90 \\ 86 \\ 261 \end{bmatrix} \pm \begin{bmatrix} 8 \\ 6 \\ 50 \end{bmatrix} \frac{N}{mm}$
X-Carriage	$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} 150 \\ 97 \\ 90 \end{bmatrix} \pm \begin{bmatrix} 9 \\ 6 \\ 5 \end{bmatrix} \frac{N}{mm}$
Full Structural Loop	$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} 70 \\ 113 \\ 17 \end{bmatrix} \pm \begin{bmatrix} 76 \\ 277 \\ 39 \end{bmatrix} \frac{N}{mm}$

As shown in the table above, the full structural loop stiffness does not meet the required specs for any axis. For both the x and y axis, the isolated spindle bracket connection appears to be the most compliant. In the z-axis, the Z-Axis Carriage + Bracket and the X-carriage when centered appear to dominate the overall compliance, based on the steps taken to isolate various components.

Experiment 1: Separation of Platform and Gantry through Ground

The first experiment was designed to determine if loads applied at the collet could be detected on the Y-carriage. If these loads are not detected, it is reasonable to assume that the clamping of the router as well as the frame of the router 'grounds out' these loads and that the dial indicator can be placed on the platform to measure the stiffness in other parts of the gantry if necessary.



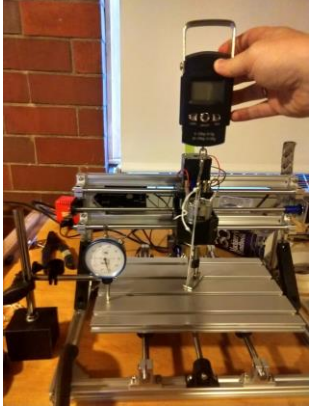
Fig 2. (A) Z deflection measurement, (B) X deflection measurement, (C) Y deflection measurement

Three experiments were performed in which the dial indicator was placed on the top surface, left side and front side of the Y carriage plate and a load between 10-20N was applied along the Z, X and Y directions respectively.

In all three the dial indicator did not move on a scale visible to the naked eye. Therefore, it is reasonable to assume that the dial indicator can be placed on the platform when measuring the stiffness of the gantry.

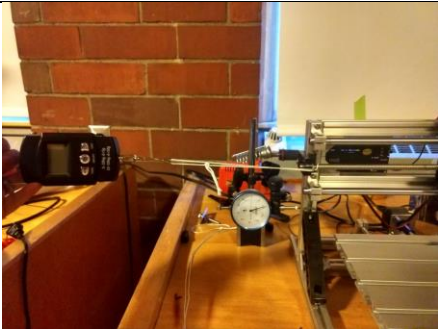
Experiment 2: Y-Platform Stiffness

The second experiment was conducted to determine the stiffness of the Y-carriage with respect to ground. Based on the geometry of the carriage, the X and Z stiffness are expected to be nearly identical and the Y axis stiffness is dependent on that of the leadscrew. The results somewhat verify these assumptions, showing that the carriage is stiffer in X-Z than Y.

Parameter	Value/Description		
Applied Force Location	Y-Platform Top surface center	Y-Platform Top surface center	Y-Platform Top surface center
Applied Force Direction	X	Y	Z
Location of dial indicator base	ground	ground	ground
Location of dial indicator tip	Y-Platform left edge	Y-Platform front edge	Y-Platform top surface
Stiffness	$1000 \pm 270 \frac{N}{mm}$	$300 \pm 16 \frac{N}{mm}$	$500 \pm 36 \frac{N}{mm}$
Photo			

Experiment 3: 8020 Gantry Cantilever Stiffness

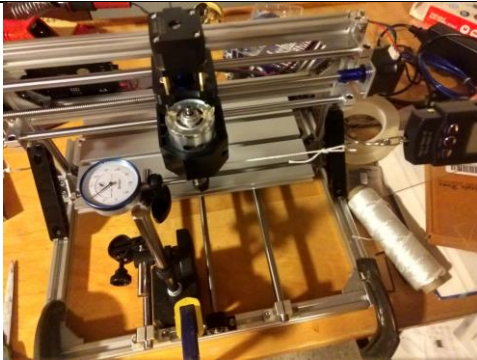
An experiment was conducted to determine the stiffness of the gantry above the plastic supporting bracket. An assumption was made in that the vertical 8020 gantry pieces below the supporting bracket mount are grounded and anything above the mount can act as a cantilever. Based on the resolution of the measurement equipment, there was no noticeable change in deflection in the z-axis and the X and Y stiffnesses are within 100N/mm of each other. A further experiment can be done next week to measure the deflection at the base of the vertical 8020 gantry to test whether it being assumed grounded is a valid assumption. Based on the measured stiffness values, this component is much stiffer than the rest of the gantry, implying that any errors due to this component will largely be from amplified abbe errors.

Parameter	Value/Description		
Applied Force Location	X-Leadscrew bearing	X-Leadscrew bearing	X-Leadscrew bearing
Applied Force Direction	X	Y	Z
Location of dial indicator base	ground	ground	ground
Location of dial indicator tip	Left 8020 plastic support on left side	Left 8020 plastic support on back side	Left 8020 plastic support on top
Stiffness	$400 \pm 74 \frac{N}{mm}$	$290 \pm 21 \frac{N}{mm}$	$Inf \frac{N}{mm}$
Photo			

Experiment 3: Full Gantry Stiffness

An experiment was conducted to measure the full gantry stiffness when a load was applied and measured at the collet. The X and Z axes were centered on their rails and the load was applied as close to the tip of the collet as possible. The deflection was measured on the flat edge of the collet as well as the plastic spindle bracket.

The low values of stiffness indicate that even if the plate on which the part was mounted was grounded, the gantry would still be out of spec.

Parameter	Value/Description		
Applied Force Location	Collet	Collet	Collet
Applied Force Direction	X	Y	Z
Location of dial indicator base	ground	Y-Carriage Platform	Y-Carriage Platform
Location of dial indicator tip	Collet flat in X-direction	Spindle Mount Bracket on Front	Spindle Mount Bracket on top
Stiffness	$90 \pm 60 \frac{N}{mm}$	$128 \pm 8 \frac{N}{mm}$	$18 \pm 3 \frac{N}{mm}$
Photo			

Experiment 4: Z-Axis Leadscrew + Spindle Bracket Stiffness

An experiment was conducted to isolate the stiffness of the z-axis leadscrew and the spindle bracket from that of the X-carriage and X-axis leadscrew. The x carriage was moved all the way to the left side to where they are in their stiffest configuration and the loads were applied and measured as close to the collet tip as possible.

These results indicate that the X-carriage stiffness along the y axis plays a significant role in the reduction of gantry stiffness in the y-direction. This makes sense due to the orientation of the rails.

Parameter	Value/Description		
Applied Force Location	Collet	Collet	Collet
Applied Force Direction	X	Y	Z
Location of dial indicator base	ground	Y-Carriage Platform	Y-Carriage Platform
Location of dial indicator tip	Spindle Mount Bracket Left Flat surface	Spindle Mount Bracket on Front	Spindle Mount Bracket on top
Stiffness	$88 \pm 6 \frac{N}{mm}$	$416 \pm 10 \frac{N}{mm}$	$67 \pm 3 \frac{N}{mm}$
Photo			

Experiment 5: Spindle Bracket Stiffness

An experiment was conducted to isolate the stiffness of the spindle bracket from the z-carriage leadscrew and rail assembly. The z-carriage was lowered all the way to its bottom position to where its rails are in their stiffest configuration. The loads were applied as close to the collet tip as possible and the results were measured on the end of the spindle bracket. These results seem somewhat counter intuitive in that I would expect the Y axis to be stiffer after removing the effects of the rails. However, this could be due to additional angular abbe errors due to the positioning of the bracket. The z axis rails play a large role in the compliance in the z-carriage+ spindle mount Z direction compliance. Because the X axis stiffness has not changed, it is reasonable to assume that the x-axis compliance comes from the spindle bracket.

Parameter	Value/Description		
Applied Force Location	Collet	Collet	Collet
Applied Force Direction	X	Y	Z
Location of dial indicator base	ground	ground	Y-Carriage Platform
Location of dial indicator tip	Spindle Mount Bracket Left Flat surface	Spindle Mount Bracket on Front	Spindle Mount Bracket on top surface
Stiffness	$90 \pm 8 \frac{N}{mm}$	$86 \pm 6 \frac{N}{mm}$	$261 \pm 50 \frac{N}{mm}$
Photo			

Experiment 6: X-Carriage Stiffness

An experiment was conducted to isolate the stiffness of the x-carriage when centered on the rails from the rest of the gantry system. The load was applied to the center of the x-carriage and measured at various points on the x-axis carriage. These low values of stiffness indicate that this carriage and rail subsystem also play a significant role in the overall compliance of the gantry.

Parameter	Value/Description		
Applied Force Location	X-Carriage center	X-Carriage center	X-Carriage center
Applied Force Direction	X	Y	Z
Location of dial indicator base	ground	ground	Y-Carriage Platform
Location of dial indicator tip	Spindle Mount Bracket Left Flat surface	Spindle Mount Bracket on Front	Spindle Mount Bracket on top surface
Stiffness	$150 \pm 9 \frac{N}{mm}$	$97 \pm 6 \frac{N}{mm}$	$90 \pm 5 \frac{N}{mm}$
Photo			

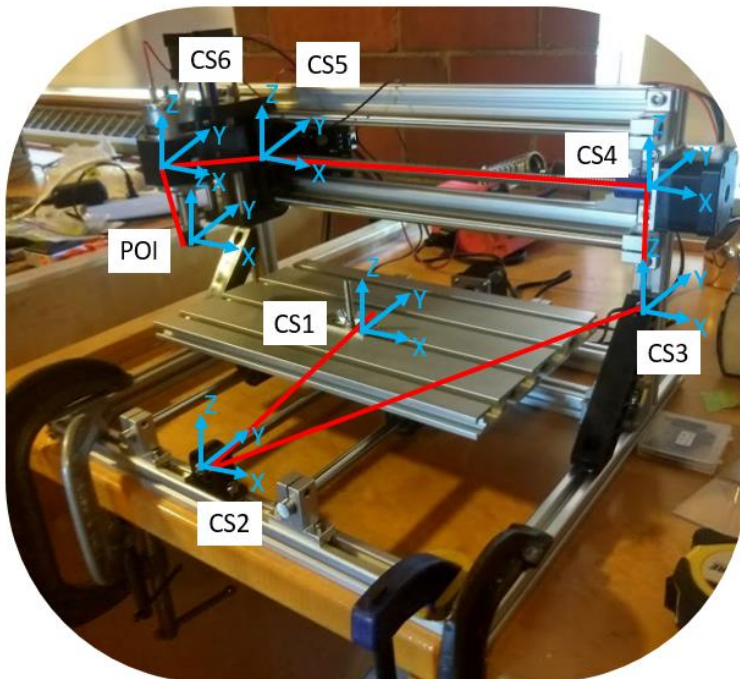
HTM Model Milestones

As detailed below, my htm model consists of 6 coordinate systems. The first coordinate system is coincident with the center of the Y-plate on the top surface. The second coordinate system is centered on the leadscrew bearing at the front of the router. The third coordinate system is centered at the top of the plastic bracket on the right side of the router. The fourth coordinate system is centered at the X-axis leadscrew coupler. The fifth coordinate system is centered at the center of the x-carriage on the leadscrew. The sixth coordinate system is centered at the edge of the plastic piece holding the motor. The point of interest is located at the tip of the engraving tool. Personally, I find it easier to align all of my axes and account for the stiffness through measurement and manipulating the coordinate systems of the beam bending analysis.

Based on my experiments, the plastic bracket + Z-carriage interface is the most compliant subsystem. In my htm assembly spreadsheet, I inputted the measured subsystem and changed the rest of the stiffness values for the assembly to be 2E11 N/mm. This results in the following:

Load Induced Errors	$\begin{bmatrix} \delta_x \\ \delta_y \\ \delta_z \end{bmatrix} = \begin{bmatrix} 113 \\ 24 \\ 149 \end{bmatrix} \mu m$
Abbe Errors	$\begin{bmatrix} \delta_x \\ \delta_y \\ \delta_z \end{bmatrix} = \begin{bmatrix} .33 \\ 2.2290 \\ .36 \end{bmatrix} \mu m$
Stiffness	$\begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} 87 \\ 415 \\ 66 \end{bmatrix} \frac{N}{mm}$

I believe that the abbe errors are so small because the rest of the structure was considered rigid and thus not 'allowed' to rotate to induce the abbe errors. Perhaps this is a flaw in my understanding of the coordinate systems or perhaps I would have seen different results if I chose different coordinate systems. For the next assessment, I believe that the abbe errors will be more prominent when I actually input the measured stiffness of the system rather than assuming rigid. However, with just modeling the most complaint component the values of stiffness are within an order of magnitude in the X and Z axis and is about four times greater in the Y axis. All in all, I would say this is a very reasonable first pass test to get a good feel for the system stiffness.



Coordinate System	Location
CS1	Top center of Y carriage Plate
CS2	Ground Frame at Y-axis leadscrew end
CS3	Top of plastic supporting brace
CS4	X-axis Lead screw end
CS5	X-axis carriage center
CS6	Tip of z-axis carriage
POI	collet/tool tip

